

Pruning Deep Neural Networks via the Marchenko–Pastur Distribution

Leonid Berlyand

Department of Mathematics
Pennsylvania State University
University Park, PA 16802, USA

Theo Bourdais

Department of Computing and Mathematical Sciences
California Institute of Technology
Pasadena, CA 91125, USA

Houman Owahdi

Department of Computing and Mathematical Sciences
California Institute of Technology
Pasadena, CA 91125, USA

Yitzchak Shmalo*

Department of Mathematics
Pennsylvania State University
University Park, PA 16802, USA

Abstract

We study a Marchenko–Pastur (MP) random-matrix approach to pruning deep neural networks with very small post-pruning fine-tuning budgets. The main practical contribution is accuracy retention under short calibration and fine-tuning schedules, rather than a long post-pruning reoptimization pipeline. The theory gives deterministic data-path certificates: if the removed component R has small propagated logit effect $L_s\|R\psi_1(s)\|_\infty$, pruning decreases an elastic-net objective and preserves samples whose dense margin exceeds twice the perturbation. The zero-budget case gives perfect pruning; a prune–restore extension models weight restoration inside a fixed sparse-execution pattern; and an additive L_2 -regularized model shows admissible random-like components vanish at the training limit, with persistent spikes stabilizing as the MP bulk collapses. Under iid-Gaussian sufficient conditions, the fitted MP edge σ_+ gives a high-probability layerwise budget signal.

On ImageNet-1k, after only three distillation epochs, ViT-B/16 2:4+ToMe reaches 83.41% top-1 (−1.70 pp from dense) at 59.81% sparse-execution MAC reduction, with $1.388\times$ best-observed A40 native-2:4 backend speedup for the same checkpoint and ToMe graph; a separate no-ToMe A100 endpoint gives $2.705\times$. At structured sparsity, ViT-B/16 6:12 reaches 83.74%, ViT-L/16 8:16 dense+permutation reaches 85.33% (−0.51 pp), and ConvNeXtV2-Base 12:16 reaches 86.35% (−0.37 pp). For CNNs, ResNet50 8:16 dense+permutation reaches 75.87% (−0.26 pp), and ResNet152d CAST-conv+permutation reaches 81.33% (−1.53 pp) at $\sim 50\%$ MAC accounting with a $1.62\times$ A40 im2col+2:4 sparse-GEMM audit.

Keywords: DNNs, ViTs, Random Matrix Theory, Marchenko–Pastur distribution, pruning, regularization

Supplementary information. Full proofs, Gaussian/random-matrix specializations, and mathematical corollaries are provided in Online Resource 1. Full methodology, run recipes, checkpoint ledgers, timing-audit details, and the complete comparison table are provided in Online Resource 2. Mirror copies of the manuscript sources and PDFs are maintained at https://github.com/yspennstate/RMT_based_pruning_in_deep_learning.

Norm convention. Throughout, $\|x\|_\infty$ denotes the vector maximum norm, while $\|A\|_\infty$ denotes the induced matrix norm

$$\|A\|_\infty := \max_i \sum_j |A_{ij}|.$$

*Corresponding author: Yitzchak Shmalo. Email: yitzchak.shmalo@gmail.com.

Contents

1	Introduction	2
2	Randomness in Deep Neural Networks	3
2.1	Introduction to Deep Neural Networks	3
2.2	The MP distribution in machine learning contexts	4
2.3	Reduction of randomness in DNN weights via MP diagnostics	4
3	Numerical evidence: RMT-guided pruning of ViTs	5
3.1	ViT-B/16 results	6
3.2	Broader architecture sweep	6
3.3	Cert-aware structured sparsity: the FLOP / speedup table	8
3.4	Comparison with published pruning and structured-sparsity baselines	11
3.5	Limitations and scope	11
4	Abstract Additive Perturbation Extensions	13
4.1	Assumptions for the generalized perturbation framework	13
4.2	Generalized perturbation and loss-reduction results	13
4.3	Main Gaussian/RMT corollaries	15
5	Main Theoretical Results: Deterministic Pruning Certificates	17
5.1	Deterministic data-path certificates for mask pruning	17

1 Introduction

DNN compression is motivated by overfitting, regularization, and deployment constraints. Random Matrix Theory (RMT) has been used to study trained spectra, implicit self-regularization, generalization diagnostics, Jacobians, and initialization [17, 14, 19, 31, 28, 22, 23, 21, 18, 16]. For fixed pruning hyperparameters, this paper uses MP spectral diagnostics to allocate masks without validation or test access at mask-construction time.

A main contribution is that the reported drops are obtained with little post-pruning fine-tuning: one epoch per unstructured pruning cycle and three distillation epochs for CAST/CAST-conv, rather than long retraining schedules. The empirical target is sparsity inside dense affine maps of Vision Transformers and related ImageNet models [5, 29]. Trained spectra are heterogeneous: the method treats MP-like layers as candidates for larger pruning budgets, while protecting non-MP or heavy-tailed layers. The main numerical results are deliberately front-loaded. On ImageNet-1k, Hybrid Magnitude-SER keeps ViT-B/16 at 83.37% top-1 at 50% unstructured sparsity, while the deployable CAST 2:4+ToMe row reaches 83.41% after only three distillation epochs at 59.81% sparse-execution MAC reduction. The same checkpoint and ToMe graph gives a measured $1.36\times$ fixed-batch A40 native-2:4 speedup ($1.388\times$ best batch-sweep value); a separate no-ToMe ViT-B/16 dense-to-2:4 A100 endpoint gives $2.705\times$. Wider structured projections improve the accuracy side: ViT-B/16 6:12 reaches 83.74%, ViT-L/16 8:16 dense+permutation reaches 85.33% (-0.51 pp), and ConvNeXtV2-Base 12:16 reaches 86.35% (-0.37 pp). For CNNs, ResNet50 8:16 dense+permutation reaches 75.87%, only 0.26 pp below dense, and ResNet152d CAST-conv+permutation reaches 81.33%, 1.53 pp below dense, with a $1.62\times$ A40 im2col+2:4 sparse-GEMM audit. The wider 6:12, 8:16, and 12:16 rows are accuracy/MAC-accounting rows, not native sparse Tensor Core throughput claims.

We also present three main theoretical results. First, the deterministic data-path certificate says that if the removed component R has small propagated logit effect $L_s\|R\psi_1(s)\|_\infty$, then pruning decreases

an elastic-net objective and preserves every training sample whose dense margin is larger than twice this perturbation (Lemma 5.1, Corollary 5.2, Theorem 5.4). In the zero-budget or “perfect pruning” case, the margin bound gives no accuracy loss on the training set. Second, the prune–restore certificate models structured $k:n$ sparsity: restoring entries inside an already-paid sparse-execution group can improve the certificate while leaving the final sparse-execution pattern unchanged (Theorem 5.5). Third, the additive L_2 -regularized theory shows that, under one-sided stationarity and local-path convergence, admissible random-like components vanish at the training limit; the associated MP bulk collapses while persistent signal spikes stabilize (Theorem 4.8, Corollaries 4.9–4.12, and the generalized Theorem 4.7). The MP edge enters as a sufficient random-matrix condition for data-path budgets and as the empirical layer-allocation signal used by SER/CAST. Tables 1, 2, 3, and 4 report the main unstructured, structured, MAC-reduction, and deployability results; Table 5 gives literature context. Prior pruning taxonomies and ViT/CNN compression citations are collected in Online Resource 2.

Section 2 gives DNN and MP preliminaries. Section 3 reports the numerical evidence. Sections 4 and 5 state the additive and deterministic certificates. Full proofs and mathematical details are in Online Resource 1; full protocols, algorithms, provenance notes, and supplementary numerical material are in Online Resource 2.

2 Randomness in Deep Neural Networks

2.1 Introduction to Deep Neural Networks

In classification tasks, the goal is to assign each element of a set S to one of K classes. Let $C(s) \in \{1, \dots, K\}$ denote the correct class of $s \in S$. Given a labeled training set $T \subset S$, we seek a classifier that generalizes from T to unseen data. We consider DNNs of the form

$$\phi(\cdot, \alpha) = \rho \circ X(\cdot, \alpha),$$

where ρ is the softmax map and $X(\cdot, \alpha)$ is a composition of affine maps and nonlinearities:

$$X(\cdot, \alpha) = \lambda \circ M_L(\cdot, \alpha) \circ \dots \circ \lambda \circ M_1(\cdot, \alpha).$$

Here:

- $M_k(\cdot, \alpha)$ is an affine map $\mathbb{R}^{N_{k-1}} \rightarrow \mathbb{R}^{N_k}$ with weight matrix $W_k \in \mathbb{R}^{N_k \times N_{k-1}}$ and bias vector $\beta_k \in \mathbb{R}^{N_k}$, so $M_k(x) = W_k x + \beta_k$.
- $\lambda : \mathbb{R}^m \rightarrow \mathbb{R}^m$ is a nonlinear activation. In the simplified theory below we take λ to be either the coordinatewise absolute value or ReLU.
- The softmax map $\rho : \mathbb{R}^K \rightarrow \mathbb{R}^K$ is given by

$$\rho(v)_i = \frac{e^{v_i}}{\sum_{j=1}^K e^{v_j}}, \quad v \in \mathbb{R}^K. \quad (1)$$

The standard cross-entropy loss is

$$L_{\text{CE}}(\alpha) = -\frac{1}{|T|} \sum_{s \in T} \log(\phi_{C(s)}(s, \alpha)). \quad (2)$$

2.2 The MP distribution in machine learning contexts

The Marchenko–Pastur distribution is a basic object in RMT [15]; high-dimensional random-matrix methods more broadly have applications in signal processing, statistics, wireless communication, and machine learning [30, 9, 26, 3]. We first define the relevant empirical spectral distributions.

Definition 2.1 (Eigenvalue and singular-value empirical spectral distributions). Let $G \in \mathbb{R}^{N \times M}$, and let $s_1(G), \dots, s_{\min\{N, M\}}(G)$ denote its singular values, counted with multiplicity and including possible zeros. The singular-value empirical spectral distribution (ESD) of G is

$$\nu_G := \frac{1}{\min\{N, M\}} \sum_{i=1}^{\min\{N, M\}} \delta_{s_i(G)}.$$

If $A \in \mathbb{R}^{M \times M}$ is symmetric positive semidefinite with eigenvalues $\lambda_1(A), \dots, \lambda_M(A)$, its eigenvalue ESD is

$$\mu_A := \frac{1}{M} \sum_{i=1}^M \delta_{\lambda_i(A)}.$$

Theorem 2.2 (Marchenko–Pastur law). *Let W_N be an $N \times M_N$ random matrix with i.i.d. entries of mean 0, variance σ^2 , and finite fourth moment. Define*

$$X_N := \frac{1}{N} W_N^\top W_N, \quad c_N := \frac{M_N}{N} \rightarrow c \in (0, \infty).$$

Then the eigenvalue ESD μ_{X_N} converges almost surely to the Marchenko–Pastur law

$$\mu_{\text{MP}}^{c, \sigma^2} = \left(1 - \frac{1}{c}\right)_+ \delta_0 + \frac{\sqrt{(\lambda_+ - x)(x - \lambda_-)}}{2\pi c \sigma^2 x} \mathbf{1}_{[\lambda_-, \lambda_+]}(x) dx, \quad (3)$$

where

$$\lambda_{\pm} = \sigma^2(1 \pm \sqrt{c})^2. \quad (4)$$

Proof. This is a classical cited random-matrix input rather than a new theorem of the paper; see Online Resource 1 for source notes.

Remark 2.3. If $M_N \leq N$ for all N , then $c \in (0, 1]$ and the atom at zero in (3) vanishes. When discussing the singular-value scale of W_N , we write

$$\sigma_+ := \sqrt{N\lambda_+}$$

for the corresponding MP upper edge.

2.3 Reduction of randomness in DNN weights via MP diagnostics

At common initialization scale W_{ij} has variance g/N , so the remark above gives $\sigma_+ = \sqrt{g}(1 + \sqrt{c})$ in the rectangular aspect-ratio limit $M/N \rightarrow c$. Singular values and the nonzero eigenvalues of $W^\top W$ are then $O(1)$, while the eigenvalues of $X_\ell = W_\ell^\top W_\ell/N$ are $O(1/N)$; learned spectra later deviate from the initial random law [17, 27]. Training introduces structure, motivating the following signal-plus-randomness suppositions.

Supposition 1. After t training steps, the layer- ℓ weight matrix can be decomposed as

$$W_\ell(t) = R_\ell(t) + S_\ell(t),$$

where $R_\ell(t)$ is an independent random perturbation and $S_\ell(t)$ is a structured signal component.

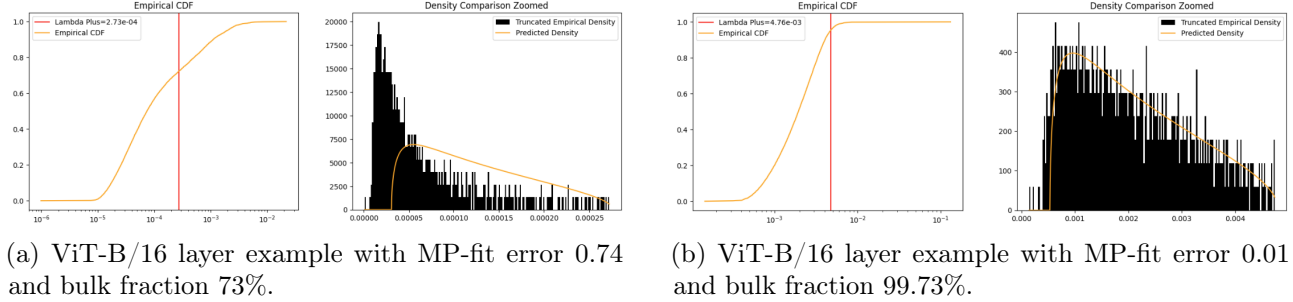


Figure 1: Layerwise MP diagnostics for two projection matrices from the same trained ViT-B/16.

Supposition 2. For the spectral spike interpretation, $S_\ell(t)$ is low rank or approximately low rank relative to the width.

These suppositions are modeling devices for the MP-budget argument, not assertions that trained ViTs are globally iid noise plus low-rank signal. Figure 1 gives two representative layer diagnostics; the assumptions behind using MP fit for budgets are stated in Sections 4 and 5.

3 Numerical evidence: RMT-guided pruning of ViTs

This section reports empirical pruning evidence and gives the minimum algorithmic context needed to read the tables. Full run recipes, scripts, checkpoint ledgers, calibration details, and migrated numerical appendices are in Online Resource 2. All ImageNet-1k top-1 values in this section are measured after the stated pruning and fine-tuning recipe on the corresponding dense checkpoint; validation labels are not used to construct masks. The dense baseline in a row is therefore part of the row’s protocol, and comparisons to external work should be read through the reported drop, compression axis, training budget, and deployment note rather than through raw top-1 alone.

The simplest baseline is magnitude pruning. For a target sparsity s , it removes the entries with the smallest absolute weights in the prunable tensors, either globally or within the layer set specified by the experiment. Magnitude pruning is useful because it is deterministic, cheap, and hard to dismiss: if an RMT-guided method cannot beat it under the same checkpoint, layer set, and fine-tuning budget, then the spectral signal has not added evidence. It is also incomplete for this setting. At high sparsity it can spend too much budget in layers whose small entries still matter along the data path, and it has no mechanism for allocating different budgets from the layer spectra.

The “classical RMT” baseline is different. It fits a Marchenko–Pastur bulk to a weight matrix or to a reshaped convolutional tensor, estimates an upper bulk edge σ_+ , and treats the sub-edge component as the random-like part suggested by the fitted spectrum. In its direct pruning form, the method removes or sparsifies this component by singular-vector reconstruction rather than by weight magnitude alone. This is an important diagnostic baseline because it asks whether the MP edge by itself identifies removable directions. Table 1 shows that it helps relative to magnitude at some higher sparsities, but it is not the final method: a literal sub-edge reconstruction is too coarse for modern ViT weights, and the paper does not claim that every sub-edge direction is noise.

The practical unstructured method is SER, short for sparsify–estimate–restore. SER starts from an over-pruned candidate, evaluates the lost mass through the same MP-guided budget view, and restores entries that are most useful for the target sparse model. The point is not to keep the smallest MP-bulk entries automatically; it is to use the MP fit as a layerwise budget signal, then let the restoration step decide which entries should return. Hybrid Magnitude–SER uses plain magnitude pruning at low sparsity, where it is strong and stable, then switches to SER once the pruning level is high enough that

layer allocation and restoration matter. In the canonical ViT-B/16 row this gives 83.37% top-1 at 50% unstructured sparsity, compared with 81.67% for magnitude and 82.57% for the direct RMT baseline under the same dense checkpoint and short post-pruning fine-tuning convention.

CAST is the structured-sparsity version of this idea. The target is a fixed $k:n$ pattern: in every group of n weights, k entries are kept and the rest are zeroed. A 2:4 pattern is the deployable NVIDIA sparse-tensor-core case; wider 6:12, 8:16, and 12:16 patterns are accuracy/MAC-accounting probes unless a native backend for that exact pattern is explicitly audited. CAST scores candidate group patterns using a certificate-inspired objective: preserve entries that matter for the propagated data-path perturbation while respecting the final group sparsity. The “free restoration” terminology means that, inside a group whose final sparse-execution cost is already fixed, restoring a kept entry can improve the certificate without changing the number of executed sparse weights.

The CAST rows should therefore be split into two categories. Rows marked 2:4 with native sparse-kernel audits are deployment evidence: the same final pattern can be represented by a backend that has a real sparse execution path. The ViT-B/16 2:4+ToMe checkpoint, for example, reaches 83.41% top-1 at 59.81% sparse-execution MAC reduction and has a measured A40 native-2:4 speedup for the same checkpoint and ToMe graph. The ResNet rows use a different audit: convolution is lowered to im2col and then timed as a 2:4 sparse GEMM endpoint, so those numbers are evidence that the weight pattern can accelerate the lowered matrix multiply, not proof of an end-to-end cuDNN Conv2d speedup. The wider $k:n$ rows answer a separate question: how much accuracy is available if the structured group is less constrained than 2:4, or if a future backend exposes a wider pattern.

Token merging is another axis and is held separate in the interpretation. ToMe reduces the number of tokens processed by the ViT graph; 2:4 pruning reduces the executed weights inside eligible linear maps. The 2:4+ToMe rows report their combined sparse-execution MAC accounting because that is the graph being evaluated, but the backend speedup claims in Table 4 specify whether ToMe is held fixed. This distinction is why the different no-ToMe A100 dense-to-2:4 endpoint is reported separately in Online Resource 2 rather than pooled with the A40 ToMe-held-fixed audits.

The result blocks have the following roles. Table 1 is the matched ViT-B/16 ablation comparing magnitude, direct RMT, SER, and Hybrid Magnitude–SER. Table 2 tests whether the Hybrid pattern survives across ViT, Swin, ConvNeXt, ResNet, and Hiera checkpoints at the same sparsity grid. Table 3 reports the structured $k:n$ rows used for the main MAC-reduction claims. Table 4 is the speed audit. The comparison table gives the closest available published contexts without treating them as a controlled leaderboard. The full theory-to-numerics map is in Online Resource 2; it matches each CAST/SER implementation step to the lemma, theorem, corollary, or empirical proxy it is meant to instantiate, and it states explicitly where the link is motivational rather than a verified certificate. The theoretical sections then explain why the algorithms are phrased as certificate-inspired pruning: the deterministic lemmas control the data-path effect of a removed component, while MP provides a sufficient random-matrix condition and a practical layerwise signal, not a complete certified transformer bound.

3.1 ViT-B/16 results

Layerwise interpretation and mask-only results are in Online Resource 2.

3.2 Broader architecture sweep

Table 2 gives the Hybrid Magnitude–SER sweep; Online Resource 2 gives architecture provenance, adaptive-variant details, slope/certificate-audit analyses, aggregate statistics, and checkpoint ledgers.

Figures 2 and 3 show the same size trend in two views: at matched nominal sparsity or MAC reduction, larger dense checkpoints tend to pay a smaller top-1 penalty. This is theoretically plausible

Table 1: ViT-B/16 pruning results on ImageNet-1k validation under this paper’s matched checkpoint, prunable-layer set, sparsity grid, evaluation pipeline, and short fine-tuning budget. Entries are top-1 accuracy (%) at sparsity s ; the dense baseline is 85.11%. Bold marks the highest entry in each sparsity column.

Method	0.05	0.10	0.15	0.20	0.25	0.30	0.35	0.40	0.45	0.50	0.55	0.60	0.65	0.70
Classical magnitude	85.21	85.16	85.08	84.87	84.65	84.46	84.08	83.53	82.86	81.67	80.17	77.98	74.30	67.44
Classical RMT	85.21	85.14	85.01	84.97	84.77	84.55	84.31	83.98	83.31	82.57	81.31	79.13	76.46	71.53
SER	84.94	84.98	84.82	84.76	84.66	84.55	84.41	84.24	83.73	83.27	82.65	81.33	79.95	77.94
Hybrid Magnitude-SER	85.23	85.17	85.06	84.89	84.81	84.64	84.55	84.28	83.80	83.37	82.76	81.39	79.95	78.01

Table 2: Hybrid Magnitude-SER results across ImageNet-1k checkpoints. Entries are post-FT top-1 accuracy (%) at sparsity s . “Base.” is the dense top-1 measured in this paper’s evaluation pipeline. Bold marks selected notable $s = 0.50$ entries: small drops from dense, high-accuracy large-model rows, or strong CNN coverage. Online Resource 2 retains the protocol markers and detailed notes for the adaptive rows.

Architecture (timm checkpoint)	Base.	0.05	0.10	0.15	0.20	0.25	0.30	0.35	0.40	0.45	0.50	0.55	0.60	0.65	0.70
ViT-B/16 augreg2_in21k_ft_in1k	85.11	85.23	85.17	85.06	84.89	84.81	84.64	84.55	84.28	83.80	83.37	82.76	81.39	79.95	78.01
ViT-B/16/384 augreg_in21k_ft	86.01	86.05	86.09	86.07	85.99	86.00	85.90	85.81	85.78	85.48	85.15	84.64	83.35	82.50	81.33
ViT-Large/16 augreg_in21k_ft	85.84	85.80	85.84	85.88	85.75	84.98	85.32	85.64	85.04	85.08	84.52	84.34	83.81	83.02	82.13
DeiT-Tiny patch16_224.fb_in1k	72.21	72.41	72.38	72.18	71.86	71.99	71.75	71.35	70.66	68.97	67.74	65.91	61.17	55.58	52.55
DeiT-Small patch16_224.fb_in1k	79.85	79.82	79.78	79.62	79.37	79.31	79.21	79.00	78.61	78.06	77.22	76.25	74.14	72.05	68.55
DeiT-Base patch16_224.fb_in1k	81.80	81.76	81.70	81.58	81.46	81.42	81.28	81.17	80.99	80.62	80.12	79.49	78.16	76.72	74.90
Swin-Tiny patch4_window7_224	81.20	81.32	81.28	81.25	81.05	81.09	80.91	80.78	80.37	80.14	79.80	78.98	78.06	76.64	74.47
ConvNeXt-Base in22k_ft_in1k	85.84	85.72	85.58	85.51	85.26	85.35	85.39	85.26	85.04	84.94	84.80	84.09	83.70	83.11	81.21
ResNet50d ra2_in1k	80.55	79.90	80.01	80.02	80.12	80.09	80.12	80.06	79.90	79.48	79.16	78.93	77.93	77.25	76.32
ResNet101d ra2_in1k	82.26	82.06	82.05	82.15	82.15	82.06	82.07	81.93	81.97	81.61	81.56	81.47	80.45	80.15	79.82
Hiera-Base+ mae_in1k_ft_in1k	84.40	85.04	84.94	84.76	84.39	77.95	83.12	82.15	82.37	82.29	82.25	82.00	80.03	80.55	78.96
ResNet18 tv_in1k	69.76	69.73	69.52	69.18	68.94	69.02	69.28	69.50	69.50	69.39	69.12	69.00	68.31	67.82	67.23
ResNet34 tv_in1k	73.28	73.00	72.67	72.39	72.62	73.06	73.09	73.16	73.16	73.00	72.88	72.74	72.12	71.59	71.44
ResNet50 tv_in1k	76.13	75.85	75.33	75.72	76.07	76.17	76.13	76.13	76.14	75.75	75.76	75.70	74.83	74.62	74.15
DeiT-Base patch16_224.fb_in1k	81.97	81.79	81.69	81.54	81.44	81.25	81.35	81.18	80.95	80.68	80.12	79.48	78.24	76.70	74.78
Swin-Tiny patch4_window7_224	81.39	81.32	81.29	81.26	81.05	80.77	80.47	80.58	80.43	80.11	79.80	78.94	78.03	76.53	74.31
ConvNeXtV2-Base fcmae_ft_in22k_in1k	86.72	86.67	86.58	86.43	86.27	85.93	85.97	85.96	85.71	85.58	85.33	84.81	84.61	83.83	81.40

for RMT-guided pruning, because the MP edge is an asymptotic spectral object that is better resolved in larger matrices, and because wider overparameterized layers can contain a larger random-like bulk reservoir whose removal has small data-path effect under the certificate model. The figures are descriptive finite-model evidence for this scaling intuition, not a theorem that size alone guarantees pruneability.

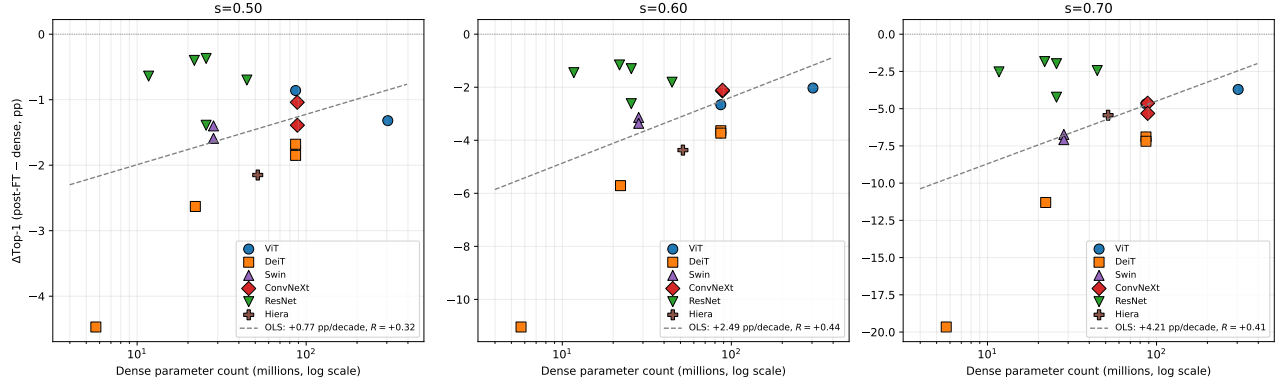


Figure 2: $\Delta\text{Top-1}$ relative to each dense baseline after Hybrid Magnitude-SER pruning vs. dense parameter count. Points are Table 2 entries; marker style encodes architecture family. Dashed lines are OLS fits at the plotted sparsity levels.

3.3 Cert-aware structured sparsity: the FLOP / speedup table

Table 3 summarizes structured sparsity; The CAST algorithm and source/checkpoint recipes are in Online Resource 2.

Table 3: CAST $k:n$ structured-projection results. “MAC red.” is dense-equivalent MAC-accounting reduction; for rows without a deploy-speed entry this is not an audited sparse-backend wall-clock claim. “Th. spd.” is $1/(1 - \text{MAC red.})$, and Δ is top-1 minus the corresponding dense baseline. “Deploy spd.” lists the selected best-observed measured sparse-backend speedup when available; Table 4 reports endpoint, batch, and audit details. Rows labeled “(ours)” are this paper’s measurements. Bold marks notable accuracy or measured-speed entries, not row ownership. A dash means no exact sparse-backend wall-clock speedup is claimed.

Architecture	Source s	Method	Dense MACs	MAC red.	Th. spd.	Deploy spd.	Top-1 (%)	Δ (pp)
<i>ViT family</i>								
ViT-B/16	0.35	Magnitude 2:4 + ToMe (ours)	17.56G	59.81%	2.49×	1.388×	82.92	−2.19
ViT-B/16	0.35	CAST 2:4 + ToMe (ours)	17.56G	59.81%	2.49×	1.388 ×	83.41	−1.70
ViT-B/16	0.35	CAST 6:12 SER+ $\alpha=0.5$ (ours, no ToMe)	17.56G	50%	2.00×	—	83.74 [§]	− 1.37
ViT-L/16	0.35	CAST 2:4 + ToMe (ours)	61.55G [†]	~60% [†]	2.50×	1.394 ×	84.37	−1.47
ViT-L/16	dense	CAST 8:16 dense+perm (ours, no ToMe)	61.55G	50%	2.00×	—	85.33 [§]	− 0.51
DeiT-B	0.35	CAST 2:4 + ToMe	17.56G	59.81%	2.49×	1.384×	80.48	−1.32
DeiT-S	0.35	CAST 2:4 + ToMe	4.61G	59.81%	2.49×	1.376×	76.96	−2.89
DeiT-T	0.35	CAST 2:4 + ToMe	1.26G	59.81%	2.49×	1.330×	65.93	−6.28
<i>ResNet family</i>								
ResNet50	0.35	CAST-conv	4.09G	48.5%	1.94×	—	73.14	−2.99
ResNet50	0.35	CAST-conv+perm (ours)	4.09G	48.5%	1.94×	1.700 ×	75.67 [‡]	− 0.46
ResNet50	dense	CAST 8:16 dense+perm (ours)	4.09G	50%	2.00×	—	75.87 [§]	− 0.26
ResNet50d	0.35	CAST-conv	4.33G	49.85%	1.99×	—	78.08	−2.47
ResNet50d	0.35	CAST-conv+perm (ours)	4.33G	49.85%	1.99×	1.496×	78.00 [‡]	−2.55
ResNet50d	dense	CAST 8:16 dense+perm (ours)	4.33G	50%	2.00×	—	78.57 [§]	−1.98
ResNet101d	0.35	CAST-conv	8.0G	~50%	2.00×	—	80.13	−2.13
ResNet101d	0.35	CAST-conv+perm (ours)	8.0G	~50%	2.00×	1.568 ×	80.59 [‡]	−1.67
ResNet101d	dense	CAST 8:16 dense+perm (ours)	8.0G	50%	2.00×	—	80.92 [§]	−1.34
ResNet152d	0.35	CAST-conv+perm (ours)	11.8G	~50%	2.00×	1.617 ×	81.33 [‡]	−1.53
<i>ConvNeXt family</i>								
ConvNeXtV2-Base	0.35	CAST 2:4 cert + free-restore (ours)	15.4G	50%	2.00×	1.295×	85.47	−1.25
ConvNeXtV2-Base	dense	CAST 12:16 dense+perm (ours, 25% sparse)	15.4G	25%	1.33×	—	86.35 [§]	− 0.37
ConvNeXtV2-Base	dense	CAST 8:16 dense+perm (ours, 50% sparse)	15.4G	50%	2.00×	—	85.85 [§]	− 0.87

[†]Analytical ViT-L/16 ToMe+2:4 estimate. [‡]Exact im2col+2:4 sparse-GEMM audit, not faster cuDNN Conv2d.

[§]Accuracy/MAC-accounting probe unless a measured endpoint is listed; dense-source wider-pattern rows are not incremental comparisons to the $s = 0.35$ source rows. *Magnitude row shares the ViT-B 2:4+ToMe deployment path; see Online Resource 2.

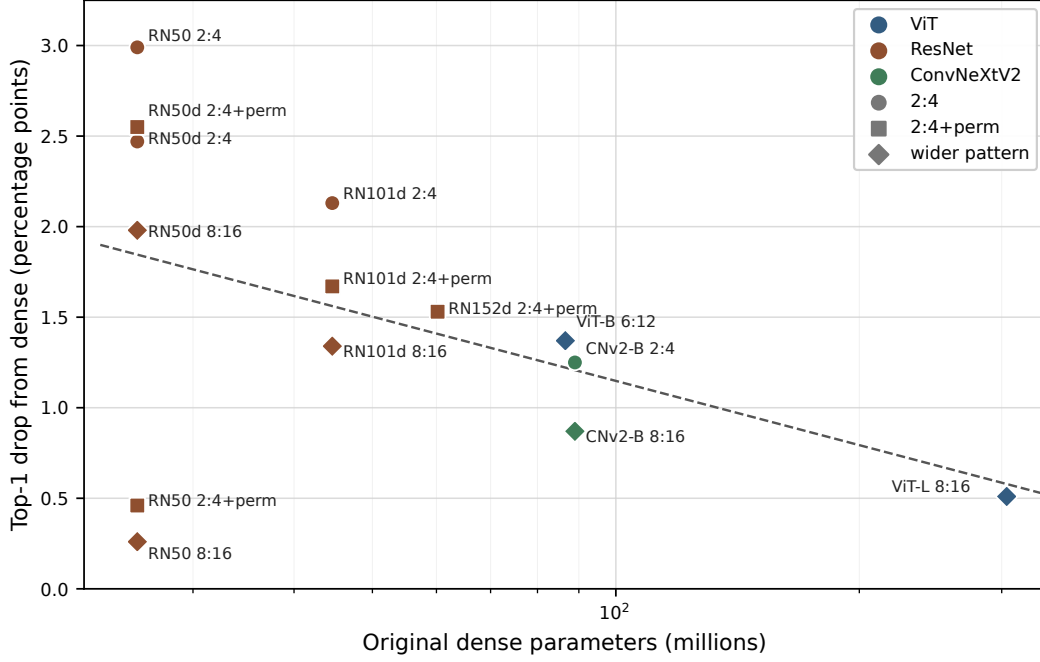


Figure 3: Top-1 drop from dense vs. original dense parameter count for rows of Table 3 at 48.5–50% dense-equivalent sparse-execution MAC reduction. The y-axis is dense top-1 minus post-FT top-1, so lower is better. The dashed line is an OLS fit in $\log_{10}$ parameters.

Table 4: Deployability audit for the main measured checkpoints, reporting best-observed A40 batch-sweep values. Accuracy is fixed by the saved post-FT checkpoint; no weights are rewritten, and only the in-memory execution endpoint changes. Linear rows use the PyTorch/NVIDIA native 2:4 backend on A40. ResNet rows use exact im2col+2:4 sparse GEMM relative to dense im2col; these rows do not claim a faster cuDNN Conv2d endpoint. Bold marks notable measured speedups, including the strongest endpoints within the Linear and ResNet audit groups.

Checkpoint	Audited endpoint	Batch	Dense im/s	Sparse im/s	Speedup	Status
ViT-B/16 + ToMe	native Linear 2:4	64	1191.8	1654.5	1.388×	native sparse Tensor Core path
ViT-L/16 + ToMe	native Linear 2:4	64	620.8	865.6	1.394×	native sparse Tensor Core path
DeiT-B + ToMe	native Linear 2:4	64	1189.6	1646.7	1.384×	native sparse Tensor Core path
DeiT-S + ToMe	native Linear 2:4	128	2708.5	3725.8	1.376×	native sparse Tensor Core path
DeiT-T + ToMe	native Linear 2:4	128	5158.4	6859.4	1.330×	native sparse Tensor Core path
ConvNeXtV2-B	native Linear 2:4	64	499.3	646.5	1.295×	pointwise Linear layers accelerated; dense convs unchanged
ResNet50	im2col sparse GEMM	128	469.5	797.9	1.700×	im2col 2:4 sparse GEMM
ResNet50d	im2col sparse GEMM	128	445.2	666.2	1.496×	im2col 2:4 sparse GEMM
ResNet101d	im2col sparse GEMM	128	228.2	357.8	1.568×	im2col 2:4 sparse GEMM
ResNet152d	im2col sparse GEMM	128	159.9	258.7	1.617×	im2col 2:4 sparse GEMM

Deployability interpretation. Endpoint coverage and the separate A100 no-ToMe ViT-B benchmark are documented in Online Resource 2.

Several CAST rows start from denser, higher-accuracy checkpoints than the cited ResNet references. In that regime the remaining weights are plausibly carrying more useful signal, so the relevant comparison is the reported drop together with the training budget and measured endpoint, not raw top-1 alone. The wider 6:12, 8:16, and 12:16 accuracy/MAC probes remain in Table 3; Table 5 carries the closest published unstructured and structured-sparsity context; the complete comparison table is

provided in Online Resource 2.

3.4 Comparison with published pruning and structured-sparsity baselines

Table 5 is literature context, not matched baseline evidence. Its main purpose is to show that the small accuracy drops reported here are obtained with very limited fine-tuning compared with many published pruning pipelines; matched evidence is Table 1, Table 2, and the caveats in Online Resource 2. We separate unstructured or fine-grained weight sparsity from deployable sparse-kernel claims: parameter sparsity and FLOP accounting are useful compression proxies, but do not by themselves imply latency or throughput gains on a given backend [7, 1, 10].

3.5 Limitations and scope

The main empirical scope qualifications are wider-pattern deployability, ResNet Conv2d endpoint support, artifact-bundle coverage, matched-ablation scope beyond ViT-B/16, and single-seed CAST row estimates; details are in Online Resource 2. The main theory limitation is the unestimated analytic L_s for modern transformer blocks; see Online Resource 1 for the local-Lipschitz discussion.

Table 5: Closest ImageNet-1k unstructured and structured-sparsity context for the main rows. This compact table keeps the most comparable representation classes from the full comparison table: unstructured/fine-grained pruning, 2:4 or $N:M$ semi-structured sparsity, and audited endpoint rows. The complete literature-context table is provided in Online Resource 2. Rows labeled “ours” are this paper’s measurements; “FT/train” is the reported training or post-pruning fine-tuning budget.

Method	Architecture	Compression	Top-1	Δ FT/train	Main comparison point
Hybrid Mag-SER (ours)	ViT-B/16	50% unstructured	83.37	-1.74 1/cycle	MP-budgeted unstructured row with short-cycle FT.
CAST 2:4+ToMe (ours)	ViT-B/16	2:4+ToMe	83.41	-1.70 3	Native A40 2:4 endpoint 1.388 $\times$; separate no-ToMe A100 endpoint 2.705 $\times$.
SNOWS [13]	ViT-B/16	2:4 QKV+Out+MLP	76.57	-3.85 0	Closest one-shot ViT 2:4 row; MiniImageNet-1k subset, no endpoint speed row.
MaskLLM-4V [6]	ViT-B/16	2:4 learned mask	79.46	+0.31 20 mask ep.	Learned-mask ViT 2:4 context; weights frozen, no ViT endpoint speed reported.
SparseFormer [8]	ViT-B/16 AugReg	latent-token reduction	83.40	-1.20 20+5 ep.	Similar accuracy but token reduction, not weight sparsity; 1.85 $\times$ throughput.
CAST 8:16 dense+perm (ours)	ViT-L/16	8:16 = 50%	85.33	-0.51 3	Main wider-pattern accuracy/MAC row; no native 8:16 endpoint audit.
ToMe [2]	ViT-L/16 MAE	token merging	85.05	-0.61 MAE FT	Closest token-merging accuracy/speed context; not weight sparsity.
UniPTS [32]	ResNet-50	50/60/70% unstructured	75.76/75.37/74.73	-0.36/ - 0.75/ - 1.39 16k iters	Closest post-training unstructured ResNet row with limited calibration.
Hybrid Mag-SER (ours)	ResNet50	50/60/70% unstructured	75.76/74.83/74.15	-0.37/ - 1.30/ - 1.98 1/cycle	This paper’s matched short-cycle unstructured ResNet sweep.
AC/DC [24]	ResNet-50	50% unstructured	77.05	+0.21 100 train	Strong global sparse-training baseline; much longer training.
Mishra et al. [20]	ResNet-50	2:4 FP16	76.20	+0.10 repeated train	Original 2:4 sparse Tensor Core reference; up to 2 $\times$ sparse math.
Pool-Yu perm. [25]	ResNet-50	2:4+perm	76.29	+0.13 90 FT	Closest channel-permutation semi-structured ResNet context.
CAST-conv+perm (ours)	ResNet152d	2:4 im2col	81.33	-1.53 3	Main CNN sparse-GEMM audit; 1.617 $\times$ A40 im2col, not cuDNN Conv2d.
CAP [12]	ConvNeXt-L CLIP	50/60/70% unstructured	87.5/87.1/86.8	-0.3/ - 0.7/ - 1.0 0	Closest high-accuracy ConvNeXt unstructured one-shot context; no endpoint speed row.
Hybrid Mag-SER (ours)	ConvNeXtV2-B	50/60/70% unstructured	85.33/84.61/81.40	-1.39/ - 2.11/ - 5.32 1/cycle	This paper’s ConvNeXtV2 unstructured row with short-cycle FT.
CAST 12:16 (ours)	ConvNeXtV2-B	$\sim$ 25% MAC-accounting	86.35	-0.37 3	Main ConvNeXtV2 wider-pattern accuracy/MAC row; no native 12:16 endpoint audit.

4 Abstract Additive Perturbation Extensions

This section gives the abstract $W = S + R$ additive analogue of the deterministic data-path framework in Section 5. Gaussian/RMT specializations and proofs are in Online Resource 1.

4.1 Assumptions for the generalized perturbation framework

Throughout the asymptotic statements in this section, the training set T is finite and fixed independently of N , unless a result explicitly states summability conditions for a growing family.

Assumption 4.1 (General architecture). Assume the DNN can be written as

$$\phi = \rho \circ \psi_2 \circ (R + S) \circ \psi_1,$$

where ψ_1, ψ_2 may depend on the remaining network parameters and ρ is softmax. For each $s \in T$, let $\mathcal{U}_s$ be any set containing the admissible interpolation segment $\{(S + \vartheta R)\psi_1(s) : \vartheta \in [0, 1]\}$. We assume that ψ_2 has a finite local Lipschitz constant with respect to ℓ_∞ on $\mathcal{U}_s$:

$$L_{\psi_2}(s) := \sup_{\substack{v, w \in \mathcal{U}_s \\ v \neq w}} \frac{\|\psi_2(v) - \psi_2(w)\|_\infty}{\|v - w\|_\infty} < \infty.$$

Assumption 4.2 (Perturbative random component). Assume that for every v measurable with respect to the conditioning variables $\sigma(S, \psi_1, \psi_2)$, including deterministic v ,

$$\mathbb{P}\left(\|Rv\|_\infty > d_1(N)J(v) \mid S, \psi_1, \psi_2\right) \leq d_2(N), \quad (5)$$

where $d_1(N), d_2(N) \rightarrow 0$ and $J(v)$ depends only on v . When a spectral interpretation is desired, we additionally assume that the singular-value ESD of R converges to a deterministic law with finite right edge $b_+ < \infty$.

For example, if $R \in \mathbb{R}^{N \times N}$ has iid $N(0, 1/N)$ entries and v is fixed conditional on S, ψ_1, ψ_2 , then a Gaussian union bound gives Assumption 4.2 with $J(v) = \|v\|_2$, $d_1(N) = 2\sqrt{2\log(2N)/N}$, and $d_2(N) = 1/N$. Gaussian/RMT specializations and proofs are in Online Resource 1.

Assumption 4.3 (Structured component for spectral conclusions). For any spectral conclusion, assume that there exists a sequence $k_N = o(\min\{N, M\})$ such that

$$\sum_{i > k_N} s_i(S)^2 \rightarrow 0.$$

The convergence is in probability when S is random.

For a DNN satisfying Assumption 4.1, define

$$h_\phi(s) := J(\psi_1(s)) L_{\psi_2}(s), \quad a_\phi(N, s) := h_\phi(s) d_1(N). \quad (6)$$

4.2 Generalized perturbation and loss-reduction results

We consider the regularized loss

$$L(\alpha(t)) = -\frac{1}{|T|} \sum_{s \in T} \log(\phi_{C(s)}(s, \alpha(t))) + \lambda_{\text{reg}} \sum_{i=1}^L \|W_i(t)\|_F^2. \quad (7)$$

Lemma 4.4. *Let*

$$X = \psi_2 \circ (R + S) \circ \psi_1(s),$$

and let α_W, α_S denote the corresponding network parameters when the relevant weight matrix is $W = R + S$ and S , respectively. Under Assumptions 4.1–4.2,

$$\mathbb{P}\left(\|X(s, \alpha_S) - X(s, \alpha_W)\|_\infty \leq d_1(N)h_\phi(s)\right) \geq 1 - d_2(N).$$

Theorem 4.5. *Assume Assumptions 4.1–4.2 and that*

$$\langle S, R \rangle_F \rightarrow 0 \quad \text{in probability as } N \rightarrow \infty.$$

Assume further that for all $s \in T$,

$$a_\phi(N, s) \rightarrow 0 \quad \text{as } N \rightarrow \infty.$$

Then removing R yields

$$L(\alpha_S) = L(\alpha_W) - \lambda_{\text{reg}}\|R\|_F^2 + o_{\mathbb{P}}(1) \quad (N \rightarrow \infty).$$

Secondary loss and finite-rank deformation corollaries, with proofs, are stated in Online Resource 1.

Theorem 4.6 (Multi-layer telescoping loss reduction). *Let $\mathcal{P}$ be a finite set of layers selected for pruning, with $|\mathcal{P}|$ fixed independently of N . For each $\ell \in \mathcal{P}$, write*

$$W_\ell = S_\ell + R_\ell,$$

and assume that, after the previous layers in $\mathcal{P}$ have already been replaced by their structured parts, the single-layer hypotheses of Theorem 4.5 continue to hold for layer ℓ with perturbation scale $a_{\phi, \ell}(N, s)$. If $\alpha^{(0)}$ denotes the original network parameters and $\alpha^{(|\mathcal{P}|)}$ the parameters after replacing every W_ℓ by S_ℓ , then

$$L(\alpha^{(|\mathcal{P}|)}) = L(\alpha^{(0)}) - \lambda_{\text{reg}} \sum_{\ell \in \mathcal{P}} \|R_\ell\|_F^2 + o_{\mathbb{P}}(1).$$

More generally, the set of pruned layers may depend on N . Let $\mathcal{P}_N$ be a sequence of layer sets, ordered in the pruning order. Assume that the corresponding one-layer logit events $E_{\ell, N}$ satisfy

$$\sum_{\ell \in \mathcal{P}_N} \mathbb{P}(E_{\ell, N}^c) \rightarrow 0,$$

that their perturbation scales are summable,

$$\sum_{\ell \in \mathcal{P}_N} \frac{1}{|T|} \sum_{s \in T} a_{\phi, \ell}(N, s) \rightarrow 0,$$

and that the accumulated cross term is negligible,

$$\sum_{\ell \in \mathcal{P}_N} \langle S_\ell, R_\ell \rangle_F = o_{\mathbb{P}}(1).$$

Then the same telescoping conclusion holds with $\mathcal{P}$ replaced by $\mathcal{P}_N$.

Theorem 4.7. *Assume Assumptions 4.1–4.2. Assume further that for all $s \in T$,*

$$a_\phi(N, s) \rightarrow 0 \quad \text{as } N \rightarrow \infty.$$

Suppose:

1. for each N , $\alpha(t, N) \rightarrow \alpha^*(N)$ in probability as $t \rightarrow \infty$;
2. the relevant limit weight matrix admits an admissible decomposition $W^*(N) = S^*(N) + R^*(N)$ satisfying

$$\langle S^*(N), R^*(N) \rangle_F \rightarrow 0 \quad \text{in probability as } N \rightarrow \infty.$$

For $\vartheta \in [0, 1]$, let $\alpha_{\vartheta}^*(N)$ denote the parameter vector obtained by replacing $W^*(N)$ with $S^*(N) + \vartheta R^*(N)$. Assume further that there exists a sequence of nonnegative random variables $\epsilon_N = o_{\mathbb{P}}(1)$ such that the one-sided directional stationarity event

$$\liminf_{h \downarrow 0} \frac{L(\alpha_{1-h}^*(N)) - L(\alpha^*(N))}{h} \geq -\epsilon_N$$

has probability tending to one. Define

$$\bar{a}_{\phi, N} := \frac{2}{|T|} \sum_{s \in T} a_{\phi}(N, s).$$

Then

$$\|R^*(N)\|_F^2 \leq \frac{\bar{a}_{\phi, N} + \epsilon_N}{2\lambda_{\text{reg}}} + o_{\mathbb{P}}(1).$$

In particular, if $\bar{a}_{\phi, N} + \epsilon_N \rightarrow 0$ in probability, then for every $\epsilon > 0$,

$$\lim_{N \rightarrow \infty} \mathbb{P}(\|R^*(N)\|_F \leq \epsilon) = 1.$$

If, in addition, there exists an admissible path decomposition

$$W(t, N) = S(t, N) + R(t, N)$$

such that for each fixed N ,

$$\|S(t, N) - S^*(N)\|_F + \|R(t, N) - R^*(N)\|_F \rightarrow 0 \quad \text{in probability as } t \rightarrow \infty,$$

and if for every fixed N and every $\epsilon > 0$,

$$\mathbb{P}(\|R^*(N)\|_F = \epsilon) = 0,$$

then

$$\lim_{N \rightarrow \infty} \lim_{t \rightarrow \infty} \mathbb{P}(\|R(t, N)\|_F \leq \epsilon) = 1.$$

4.3 Main Gaussian/RMT corollaries

The following corollaries record the Gaussian/RMT specialization used in the MP interpretation. They are stated here because they are main mathematical consequences of the framework; their proofs and the detailed Gaussian assumptions are in Online Resource 1.

Theorem 4.8 (Gaussian stationarity collapse). *Consider the square iid-Gaussian specialization of the additive framework for a trained limit layer*

$$W_2^*(N) = S_2^*(N) + R_2^*(N), \quad R_2^*(N)_{ij} \sim N\left(0, \frac{g_*(N)}{N}\right),$$

with the same local-path, cross-term, and one-sided stationarity hypotheses as Theorem 4.7. If the propagated perturbation scale and stationarity error vanish, then

$$\|R_2^*(N)\|_F \rightarrow 0 \quad \text{in probability.}$$

If an admissible training-path decomposition converges locally to this limit decomposition, then for every $\epsilon > 0$,

$$\lim_{N \rightarrow \infty} \lim_{t \rightarrow \infty} \mathbb{P}(\|R_2(t, N)\|_F \leq \epsilon) = 1.$$

Proof. See Online Resource 1, proof of the Gaussian stationarity-collapse theorem.

Corollary 4.9 (Variance-scale collapse at the limit point). *Under the hypotheses of Theorem 4.8, the Gaussian variance scale satisfies*

$$g_*(N)N \rightarrow 0.$$

Consequently, in the square case the MP singular-value edge of the admissible limit perturbation satisfies

$$\sigma_{+,*}^{\text{bulk}}(N) = 2\sqrt{g_*(N)} \rightarrow 0.$$

For a rectangular $N \times M_N$ perturbation with entries $N(0, g_*(N)/N)$, the corresponding Frobenius collapse condition is $g_*(N)M_N \rightarrow 0$, equivalent to $g_*(N)N \rightarrow 0$ when $M_N/N \rightarrow c \in (0, \infty)$.

Corollary 4.10 (Bulk collapse and spike stabilization at the limit point). *Under the hypotheses of Theorem 4.8, assume the structured component is spectrally admissible, with singular values $\sigma_i^*(N)$ and approximate rank scale $k_N = o(N)$. If $s_i(W_2^*(N))$ are the singular values of the full layer, then $\|R_2^*(N)\|_{\text{op}} \rightarrow 0$ in probability, each fixed spike satisfies*

$$|s_i(W_2^*(N)) - \sigma_i^*(N)| \rightarrow 0 \quad \text{in probability,}$$

the tail energy $\sum_{i > k_N} s_i(W_2^*(N))^2 \rightarrow 0$ in probability, and the singular-value empirical distribution has no mass above any fixed $\epsilon > 0$ asymptotically. If $\sigma_i^*(N) \rightarrow \bar{\sigma}_i$, then $s_i(W_2^*(N)) \rightarrow \bar{\sigma}_i$ for each fixed i .

Corollary 4.11 (Singular-subspace stabilization under an asymptotic gap condition). *Under the hypotheses of Theorem 4.8, fix $k \geq 1$ and let $\delta_k(N) = \sigma_k^*(N) - \sigma_{k+1}^*(N)$. If $\mathbb{P}(\delta_k(N) > 0) \rightarrow 1$ and*

$$\frac{\|R_2^*(N)\|_{\text{op}}}{\delta_k(N)} \rightarrow 0 \quad \text{in probability,}$$

then the top- k left and right singular-subspace projectors of $W_2^*(N)$ converge in operator norm to those of $S_2^*(N)$.

Corollary 4.12 (Eventual Gaussian supercriticality of persistent spikes). *Under the hypotheses of Theorem 4.8, assume a square finite-rank Gaussian deformation regime and fixed persistent spikes*

$$\sigma_i^*(N) \rightarrow \bar{\sigma}_i > 0, \quad 1 \leq i \leq k.$$

Then $\mathbb{P}(\sigma_i^*(N) > \sqrt{g_*(N)}) \rightarrow 1$, so these spikes are eventually supercritical above the Gaussian bulk edge $2\sqrt{g_*(N)}$. With the usual spike-separation condition, the corresponding spectral projectors of $W_2^*(N)$ converge to those of $S_2^*(N)$, and the inverse-BBP estimator

$$\hat{\sigma}_i^{\text{BBP}}(N) := \frac{1}{2} \left(s_i(W_2^*(N)) + \sqrt{s_i(W_2^*(N))^2 - 4g_*(N)} \right)$$

is consistent in probability.

5 Main Theoretical Results: Deterministic Pruning Certificates

This section states deterministic data-path certificates based on $B_T(R)$. The mask-specific results concern fixed kept/removed supports and do not assume an additive Gaussian decomposition; random models only supply sufficient bounds for that budget. Proofs are collected in Online Resource 1.

5.1 Deterministic data-path certificates for mask pruning

Fix one target layer $A \in \mathbb{R}^{n \times m}$ inside a network and write the logits as

$$z_A(s) := \psi_2(A\psi_1(s)) \in \mathbb{R}^K, \quad s \in T,$$

where ψ_1 and ψ_2 denote the pre-layer and post-layer computations with all other parameters frozen. For a decomposition

$$A = S + R, \quad A_\theta := S + \theta R, \quad 0 \leq \theta \leq 1,$$

let L_s be any finite deterministic upper bound on the local ℓ_∞ -Lipschitz constant of ψ_2 on the path

$$\{A_\theta\psi_1(s) : 0 \leq \theta \leq 1\}.$$

Define the data-path budget of the removed component by

$$B_T(R) := \frac{2}{|T|} \sum_{s \in T} L_s \|R\psi_1(s)\|_\infty. \quad (8)$$

If L_s is itself estimated from the realized network, the statements below are deterministic conditional on that realized upper bound.

Lemma 5.1 (Deterministic cross-entropy path bound). *For every decomposition $A = S + R$ and every $\theta \in [0, 1]$,*

$$|\mathcal{L}_{\text{CE}}(A_\theta) - \mathcal{L}_{\text{CE}}(A_1)| \leq (1 - \theta)B_T(R),$$

where

$$\mathcal{L}_{\text{CE}}(A) := \frac{1}{|T|} \sum_{s \in T} \left[-z_A(s)_{C(s)} + \log \left(\sum_{j=1}^K e^{z_A(s)_j} \right) \right].$$

Corollary 5.2 (Deterministic margin stability). *Let*

$$\eta_s(\theta) := (1 - \theta)L_s \|R\psi_1(s)\|_\infty.$$

If

$$p_A(s) \in \arg \max_j z_A(s)_j, \quad \Delta_A^{\text{pred}}(s) := z_A(s)_{p_A(s)} - \max_{j \neq p_A(s)} z_A(s)_j,$$

then the predicted label at s can change between A_1 and A_θ only if

$$\Delta_{A_1}^{\text{pred}}(s) \leq 2\eta_s(\theta).$$

For the true-label margin

$$\Delta_A^{\text{true}}(s) := z_A(s)_{C(s)} - \max_{j \neq C(s)} z_A(s)_j,$$

one has

$$|\text{acc}_{A_\theta}(T) - \text{acc}_{A_1}(T)| \leq \frac{1}{|T|} \# \{s \in T : |\Delta_{A_1}^{\text{true}}(s)| \leq 2\eta_s(\theta)\}.$$

Here

$$\text{acc}_A(T) := \frac{1}{|T|} \# \{s \in T : \Delta_A^{\text{true}}(s) > 0\}.$$

Theorem 5.3 (Deterministic additive Frobenius certificate). *For any decomposition $A = S + R$, not necessarily a mask decomposition, define the one-layer Frobenius-regularized objective*

$$\mathcal{J}_2(A) := \mathcal{L}_{\text{CE}}(A) + \lambda_2 \|A\|_F^2 + \mathcal{C}, \quad \lambda_2 \geq 0,$$

where $\mathcal{C}$ contains all terms independent of A . Then for every $\theta \in [0, 1]$,

$$\mathcal{J}_2(A_\theta) - \mathcal{J}_2(A_1) \leq (1 - \theta)B_T(R) - \lambda_2(1 - \theta^2)\|R\|_F^2 - 2\lambda_2(1 - \theta)\langle S, R \rangle_F.$$

Consequently, if $|\langle S, R \rangle_F| \leq \eta$, then

$$\mathcal{J}_2(A_\theta) - \mathcal{J}_2(A_1) \leq (1 - \theta)B_T(R) - \lambda_2(1 - \theta^2)\|R\|_F^2 + 2\lambda_2(1 - \theta)\eta.$$

In particular, full removal is certified to decrease the Frobenius-regularized objective whenever

$$B_T(R) + 2\lambda_2|\langle S, R \rangle_F| < \lambda_2\|R\|_F^2.$$

Theorem 5.4 (Deterministic elastic-net mask pruning certificate). *Assume $A = S + R$ is a mask decomposition, meaning*

$$S \odot R = 0.$$

For constants $\lambda_2, \lambda_1 \geq 0$, define the one-layer elastic-net objective by

$$\mathcal{J}(A) := \mathcal{L}_{\text{CE}}(A) + \lambda_2 \|A\|_F^2 + \lambda_1 \|A\|_1 + \mathcal{C},$$

where $\mathcal{C}$ contains all terms independent of A . Then for every $\theta \in [0, 1]$,

$$\mathcal{J}(A_\theta) - \mathcal{J}(A_1) \leq (1 - \theta)B_T(R) - \lambda_2(1 - \theta^2)\|R\|_F^2 - \lambda_1(1 - \theta)\|R\|_1.$$

For full pruning,

$$\mathcal{J}(S) \leq \mathcal{J}(A) - \lambda_2\|R\|_F^2 - \lambda_1\|R\|_1 + B_T(R).$$

Consequently, the mask is certified to decrease the elastic-net objective whenever

$$B_T(R) < \lambda_2\|R\|_F^2 + \lambda_1\|R\|_1.$$

Theorem 5.5 (Deterministic prune-restore certificate). *Let the original layer decompose as*

$$A = S + Q + P, \quad S \odot Q = S \odot P = Q \odot P = 0.$$

Interpret P as the entries finally removed and Q as entries restored, or kept, from an over-pruned reservoir. Define

$$A_\theta := S + Q + \theta P, \quad 0 \leq \theta \leq 1.$$

For the elastic-net objective $\mathcal{J}$ of Theorem 5.4, one has

$$\mathcal{J}(A_\theta) - \mathcal{J}(A) \leq (1 - \theta)B_T(P) - \lambda_2(1 - \theta^2)\|P\|_F^2 - \lambda_1(1 - \theta)\|P\|_1.$$

For the final prune-restore layer $S + Q$,

$$\mathcal{J}(S + Q) \leq \mathcal{J}(A) - \lambda_2\|P\|_F^2 - \lambda_1\|P\|_1 + B_T(P).$$

Consequently, the final SER-style mask is certified to decrease the elastic-net objective whenever

$$B_T(P) < \lambda_2\|P\|_F^2 + \lambda_1\|P\|_1.$$

Corollary 5.6 (When restoration improves the certificate upper bound). *Under the hypotheses of Theorem 5.5, set $R := Q + P$. Comparing the all-pruned certificate for $A = S + R$ with the restore- Q certificate for $A = (S + Q) + P$, the restored upper bound is strictly smaller iff*

$$B_T(R) - B_T(P) > \lambda_2 \|Q\|_F^2 + \lambda_1 \|Q\|_1.$$

Coordinatewise weighted variants are stated in Online Resource 1.

Theorem 5.7 (One-sided stationarity controls removable masks). *Under the hypotheses of Theorem 5.4, assume the one-sided directional stationarity condition holds for some $\varepsilon \geq 0$:*

$$\liminf_{h \downarrow 0} \frac{\mathcal{J}(A_{1-h}) - \mathcal{J}(A_1)}{h} \geq -\varepsilon.$$

Then

$$2\lambda_2 \|R\|_F^2 + \lambda_1 \|R\|_1 \leq B_T(R) + \varepsilon.$$

More generally, if

$$\mathcal{J}(A_\theta) \geq \mathcal{J}(A_1) - \varepsilon(1 - \theta) \quad \forall \theta \in [0, 1],$$

then for every $\theta < 1$,

$$\lambda_2(1 + \theta) \|R\|_F^2 + \lambda_1 \|R\|_1 \leq B_T(R) + \varepsilon.$$

Theorem 5.8 (Deterministic multi-layer mask certificate). *Let $\mathcal{J}(\alpha)$ denote the full cross-entropy plus layerwise elastic-net objective. Consider an ordered finite sequence of mask-pruning operations on layers $j = 1, \dots, M$. At step j , write the current layer after steps $1, \dots, j - 1$ as*

$$A_j = S_j + R_j, \quad S_j \odot R_j = 0,$$

and let $B_j(R_j)$ be its data-path budget in that intermediate network. If $\alpha^{(0)}$ and $\alpha^{(M)}$ denote the parameters before and after the M mask removals, then

$$\mathcal{J}(\alpha^{(M)}) \leq \mathcal{J}(\alpha^{(0)}) - \sum_{j=1}^M \lambda_{2,j} \|R_j\|_F^2 - \sum_{j=1}^M \lambda_{1,j} \|R_j\|_1 + \sum_{j=1}^M B_j(R_j),$$

where $\lambda_{2,j}$ and $\lambda_{1,j}$ are the elastic-net weights applied to layer j .

The corresponding multi-layer margin-stability corollary is stated in Online Resource 1.

Theorem 5.9 (Multi-layer stationarity controls jointly removable masks). *Use the notation of Theorem 5.8. For $h \in [0, 1]$, let α_h be obtained by applying the same ordered pruning operations with each current layer $A_j = S_j + R_j$ replaced by $S_j + (1 - h)R_j$. Assume that each $B_j(R_j)$ remains valid along this fractional path, uniformly for $h \in [0, 1]$, so the step- j cross-entropy increase is at most $hB_j(R_j)$. If the one-sided directional stationarity condition*

$$\liminf_{h \downarrow 0} \frac{\mathcal{J}(\alpha_h) - \mathcal{J}(\alpha^{(0)})}{h} \geq -\varepsilon$$

holds for some $\varepsilon \geq 0$, then

$$2 \sum_{j=1}^M \lambda_{2,j} \|R_j\|_F^2 + \sum_{j=1}^M \lambda_{1,j} \|R_j\|_1 \leq \sum_{j=1}^M B_j(R_j) + \varepsilon.$$

More generally, if

$$\mathcal{J}(\alpha_h) \geq \mathcal{J}(\alpha^{(0)}) - \varepsilon h \quad \forall h \in [0, 1],$$

then for every $h \in (0, 1]$,

$$\sum_{j=1}^M \lambda_{2,j}(2-h)\|R_j\|_F^2 + \sum_{j=1}^M \lambda_{1,j}\|R_j\|_1 \leq \sum_{j=1}^M B_j(R_j) + \varepsilon.$$

Corollary 5.10 (Gaussian sufficient condition for the data-path budget). *Let $R \in \mathbb{R}^{n \times m}$ have independent entries $R_{ij} \sim N(0, g/n)$, independent of the collection $v_s := \psi_1(s)$, $s \in T$. Assume L_s are nonnegative upper bounds for the post-layer Lipschitz constants, deterministic or measurable with respect to variables independent of R . Then for every finite T and every $\delta \in (0, 1)$, with probability at least $1 - \delta$,*

$$B_T(R) \leq \frac{2}{|T|} \sum_{s \in T} L_s \|v_s\|_2 \sqrt{\frac{2g}{n} \log \frac{2n|T|}{\delta}}.$$

Corollary 5.11 (MP-edge form of the Gaussian data-path budget). *Under the hypotheses of Corollary 5.10, define the nominal rectangular MP singular-value edge of $R \in \mathbb{R}^{n \times m}$ by*

$$\sigma_{+,R} := \sqrt{g} \left(1 + \sqrt{\frac{m}{n}} \right).$$

Then, with probability at least $1 - \delta$,

$$B_T(R) \leq \frac{2\sigma_{+,R}}{1 + \sqrt{m/n}} \sqrt{\frac{2}{n} \log \frac{2n|T|}{\delta}} \frac{1}{|T|} \sum_{s \in T} L_s \|v_s\|_2.$$

Sub-Gaussian and simultaneous multi-layer budget corollaries are stated in Online Resource 1.

Acknowledgments

LB, HO, and YS acknowledge support from NASA via the AIST program (Kernel Flows: Emulating Complex Models for Massive Data Sets). This work started during LB's sabbatical stay at Caltech hosted by H. Owahdi; LB and YS are grateful for the hospitality during their NASA-supported visit.

Statements and Declarations

Funding

The work of LB was partially supported by NSF grant DMS-2005262 and NSF grant IMPRESS-U 2401227. TB and HO acknowledge support from the Air Force Office of Scientific Research under MURI award number FA9550-20-1-0358 (Machine Learning and Physics-Based Modeling and Simulation) and FOA-AFRL-AFOSR-2023-0004 (Mathematics of Digital Twins), and from the Department of Energy under award number DE-SC0023163 (SEA-CROGS: Scalable, Efficient, and Accelerated Causal Reasoning Operators, Graphs and Spikes for Earth and Embedded Systems). HO and TB also acknowledge support from the DoD Vannevar Bush Faculty Fellowship Program under award number ONR-N000142512035. The work of YS was partially supported by the European Research Council under the European Union's Horizon 2022 research and innovation program (grant agreement No. 101041711), the Simons Foundation, the Israel Science Foundation (grant number 2258/19), and the Israel Science Foundation (ISF Grant 4101/25).

Competing interests

The authors declare that they have no competing interests relevant to the content of this article.

Author contributions

All authors contributed to the mathematical development, interpretation of results, manuscript revision, and approval of the submitted version.

Data, code, and supplementary information availability

Code, plotting scripts, pruning recipes, manuscript artifacts, and mirror copies of Online Resource 1 and Online Resource 2 are available at https://github.com/yspennstate/RMT_based_pruning_in_deep_learning. The checkpoints, validation ledgers, sparse-backend timing evidence, and per-run files supporting the main ImageNet-1k tables are available at <https://drive.google.com/drive/folders/1mm990SHAH1YdISHxirvMRdVQEAjpIxDb>. ImageNet-1k images are not redistributed; ImageNet is identified by [4], and users must obtain the dataset under the official download terms [11]. Online Resource 2 identifies the source checkpoints, recipes, evidence files, timing audits, and complete numerical ledgers for the reported rows.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author(s) used generative AI tools to accelerate drafting and revision. All mathematical claims, proofs, numerical interpretations, and bibliographic information were subsequently reviewed and edited by the author(s), who take full responsibility for the content of the manuscript.

References

- [1] Davis Blalock, Jose Javier Gonzalez Ortiz, Jonathan Frankle, and John Gutttag. What is the state of neural network pruning? In *Proceedings of Machine Learning and Systems*, volume 2, pages 129–146, 2020. URL https://proceedings.mlsys.org/paper_files/paper/2020/hash/6c44dc73014d66ba49b28d483a8f8b0d-Abstract.html.
- [2] Daniel Bolya, Cheng-Yang Fu, Xiaoliang Dai, Peizhao Zhang, Christoph Feichtenhofer, and Judy Hoffman. Token merging: Your ViT but faster. In *International Conference on Learning Representations*, 2023. URL <https://openreview.net/forum?id=JroZRarW7Eu>.
- [3] Romain Couillet and Mérouane Debbah. *Random Matrix Methods for Wireless Communications*. Cambridge University Press, 2011. doi: 10.1017/CBO9780511994746. URL <https://doi.org/10.1017/CBO9780511994746>.
- [4] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. Imagenet: A large-scale hierarchical image database. In *2009 IEEE Conference on Computer Vision and Pattern Recognition*, pages 248–255, 2009. doi: 10.1109/CVPR.2009.5206848.
- [5] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations*, 2021.

- [6] Gongfan Fang, Hongxu Yin, Saurav Muralidharan, Greg Heinrich, Jeff Pool, Jan Kautz, Pavlo Molchanov, and Xinchao Wang. MaskLLM: Learnable semi-structured sparsity for large language models. In *Advances in Neural Information Processing Systems*, volume 37, pages 7736–7758. Curran Associates, Inc., 2024. doi: 10.52202/079017-0248. URL https://proceedings.neurips.cc/paper_files/paper/2024/file/0e9a05f5ce62284c91e4a33498899124-Paper-Conference.pdf.
- [7] Trevor Gale, Erich Elsen, and Sara Hooker. The state of sparsity in deep neural networks, 2019. URL <https://arxiv.org/abs/1902.09574>.
- [8] Ziteng Gao, Zhan Tong, Kevin Qinghong Lin, Joya Chen, and Mike Zheng Shou. Bootstrapping SparseFormers from vision foundation models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 17710–17721, June 2024. doi: 10.1109/CVPR52733.2024.01677. URL https://openaccess.thecvf.com/content/CVPR2024/html/Gao_Bootstrapping_SparseFormers_from_Vision_Foundation_Models_CVPR_2024_paper.html.
- [9] Jungang Ge, Ying-Chang Liang, Zhidong Bai, and Guangming Pan. Large-dimensional random matrix theory and its applications in deep learning and wireless communications. *Random Matrices: Theory and Applications*, 10(4):2230001, 2021. doi: 10.1142/S2010326322300017. URL <https://doi.org/10.1142/S2010326322300017>.
- [10] Torsten Hoefer, Dan Alistarh, Tal Ben-Nun, Nikoli Dryden, and Alexandra Peste. Sparsity in deep learning: Pruning and growth for efficient inference and training in neural networks. *Journal of Machine Learning Research*, 22(241):1–124, 2021. URL <https://www.jmlr.org/papers/v22/21-0366.html>.
- [11] ImageNet. ImageNet download and terms. <https://www.image-net.org/download.php>, 2026. Accessed 2026-05-13.
- [12] Denis Kuznedelev, Eldar Kurtić, Elias Frantar, and Dan Alistarh. CAP: Correlation-aware pruning for highly-accurate sparse vision models. In *Advances in Neural Information Processing Systems*, volume 36, pages 28805–28831. Curran Associates, Inc., 2023. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/5bd9fbb3a5a985f80c16ddd0ec1dfc43-Abstract-Conference.html.
- [13] Ryan Lucas and Rahul Mazumder. Preserving deep representations in one-shot pruning: A hessian-free second-order optimization framework. In *International Conference on Learning Representations*, 2025. doi: 10.48550/arXiv.2411.18376. URL <https://openreview.net/forum?id=eNQp79A50z>.
- [14] Michael W. Mahoney and Charles H. Martin. Traditional and heavy tailed self regularization in neural network models. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 4284–4293. PMLR, 2019. URL <https://proceedings.mlr.press/v97/mahoney19a.html>.
- [15] V. A. Marchenko and L. A. Pastur. Distribution of eigenvalues for some sets of random matrices. *Mathematics of the USSR-Sbornik*, 1(4):457–483, 1967. doi: 10.1070/SM1967v001n04ABEH001994. URL <https://www.mathnet.ru/eng/sm4101>.
- [16] Charles H. Martin and Michael W. Mahoney. Heavy-tailed universality predicts trends in test accuracies for very large pre-trained deep neural networks. In *Proceedings of the 2020 SIAM International Conference on Data Mining*, pages 505–513, 2020. doi: 10.1137/1.9781611976236.57. URL <https://epubs.siam.org/doi/10.1137/1.9781611976236.57>.

- [17] Charles H. Martin and Michael W. Mahoney. Implicit self-regularization in deep neural networks: Evidence from random matrix theory and implications for learning. *Journal of Machine Learning Research*, 22(165):1–73, 2021. URL <http://jmlr.org/papers/v22/20-410.html>.
- [18] Charles H. Martin, Tongsu (Serena) Peng, and Michael W. Mahoney. Predicting trends in the quality of state-of-the-art neural networks without access to training or testing data. *Nature Communications*, 12(1):4122, 2021. doi: 10.1038/s41467-021-24025-8. URL <https://doi.org/10.1038/s41467-021-24025-8>.
- [19] Xuran Meng and Jianfeng Yao. Impact of classification difficulty on the weight matrices spectra in deep learning and application to early-stopping. *Journal of Machine Learning Research*, 24(28):1–40, 2023. URL <http://jmlr.org/papers/v24/21-1441.html>.
- [20] Asit Mishra, Jorge Albericio Latorre, Jeff Pool, Darko Stosic, Dusan Stosic, Ganesh Venkatesh, Chong Yu, and Paulius Micikevicius. Accelerating sparse deep neural networks. *arXiv preprint arXiv:2104.08378*, 2021. doi: 10.48550/arXiv.2104.08378.
- [21] Thiziri Nait Saada and Jared Tanner. On the initialisation of wide low-rank feedforward neural networks. *arXiv preprint arXiv:2301.13710*, 2023. doi: 10.48550/arXiv.2301.13710. URL <https://arxiv.org/abs/2301.13710>.
- [22] Leonid Pastur. On random matrices arising in deep neural networks: Gaussian case. *Pure and Applied Functional Analysis*, 5(6):1395–1424, 2020. URL <https://arxiv.org/abs/2001.06188>.
- [23] Leonid Pastur and Victor Slavin. On random matrices arising in deep neural networks: General I.I.D. case. *Random Matrices: Theory and Applications*, 12(1):2250046, 2023. doi: 10.1142/S2010326322500460. URL <https://doi.org/10.1142/S2010326322500460>.
- [24] Alexandra Peste, Eugenia Iofinova, Adrian Vladu, and Dan Alistarh. AC/DC: Alternating compressed/decompressed training of deep neural networks. In *Advances in Neural Information Processing Systems*, volume 34, pages 8557–8570. Curran Associates, Inc., 2021. URL https://proceedings.neurips.cc/paper_files/paper/2021/hash/48000647b315f6f00f913caa757a70b3-Abstract.html.
- [25] Jeff Pool and Chong Yu. Channel permutations for N:M sparsity. In *Advances in Neural Information Processing Systems*, volume 34, pages 13316–13327, 2021. URL <https://proceedings.neurips.cc/paper/2021/hash/6e8404c3b93a9527c8db241a1846599a-Abstract.html>.
- [26] V. Serdobolskii. *Multivariate Statistical Analysis: A High-Dimensional Approach*. Springer Dordrecht, 2000. doi: 10.1007/978-94-015-9468-4. URL <https://link.springer.com/book/10.1007/978-94-015-9468-4>.
- [27] Max Staats, Matthias Thamm, and Bernd Rosenow. Boundary between noise and information applied to filtering neural network weight matrices. *Physical Review E*, 108(2):L022302, 2023. doi: 10.1103/PhysRevE.108.L022302.
- [28] Matthias Thamm, Max Staats, and Bernd Rosenow. Random matrix analysis of deep neural network weight matrices. *Physical Review E*, 106(5):054124, 2022. doi: 10.1103/PhysRevE.106.054124. URL <https://doi.org/10.1103/PhysRevE.106.054124>.
- [29] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, volume 30, pages 5998–6008, 2017. URL https://papers.nips.cc/paper_files/paper/2017/hash/3f5ee243547dee91fbd053c1c4a845aa-Abstract.html.

- [30] Roman Vershynin. *High-Dimensional Probability: An Introduction with Applications in Data Science*. Cambridge University Press, 2018. doi: 10.1017/9781108231596. URL <https://doi.org/10.1017/9781108231596>.
- [31] Xuanzhe Xiao, Zeng Li, Chuanlong Xie, and Fengwei Zhou. Heavy-tailed regularization of weight matrices in deep neural networks. In *Artificial Neural Networks and Machine Learning – ICANN 2023*, volume 14263 of *Lecture Notes in Computer Science*, pages 236–247. Springer, Cham, 2023. doi: 10.1007/978-3-031-44204-9_20.
- [32] Jingjing Xie, Yuxin Zhang, Mingbao Lin, Zhihang Lin, Liujuan Cao, and Rongrong Ji. UniPTS: A unified framework for proficient post-training sparsity. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 5746–5755, June 2024. doi: 10.1109/CVPR52733.2024.00549. URL https://openaccess.thecvf.com/content/CVPR2024/html/Xie_UniPTS_A_Unified_Framework_for_Proficient_Post-Training_Sparsity_CVPR_2024_paper.html.